Diagnostic Reference Guide

Quick Reference for Laboratory Values and Imaging Findings

MediAssist Health Network
Central Laboratory & Radiology
Document ref: LAB-DIAG-007 · Version 6.1
Access: Doctors and Admin

This guide supports rapid interpretation of common laboratory values and imaging findings. Reference ranges follow MediAssist central laboratory standards; critical values trigger the notification protocol in the final section.

1. Haematology Reference Ranges

Test	Normal Range	Critical Value	Action
Haemoglobin (male)	13–17 g/dL	< 7 g/dL	Activate transfusion protocol
Haemoglobin (female)	12–15 g/dL	< 7 g/dL	Activate transfusion protocol
WBC count	4–11 × 10^9/L	> 30 × 10^9/L	Haematology consult
Platelet count	150–400 × 10^9/L	< 50 × 10^9/L	Bleeding-risk protocol
PT / INR	INR 0.9–1.2	INR > 5	Withhold anticoagulant; review
aPTT	26–36 s	> 90 s	Review heparin; bleeding risk
D-dimer	< 0.5 µg/mL	Marked rise	Consider VTE; correlate clinically
Ferritin	30–300 ng/mL	—	Low: iron deficiency; high: inflammation
ESR	M < 15, F < 20 mm/hr	—	Non-specific inflammation marker

2. Biochemistry Reference Ranges

Test	Normal Range	Critical / Significant	Note
Sodium	135–145 mEq/L	< 120 or > 160	Correct slowly
Potassium	3.5–5.0 mEq/L	< 3.0 or > 6.0	Cardiac monitoring if extreme
Creatinine	0.6–1.3 mg/dL	Rapid rise	Assess AKI; review nephrotoxics
Urea	15–45 mg/dL	—	Correlate with creatinine
Glucose (fasting)	70–100 mg/dL	< 50 or > 400	Treat hypo-/hyperglycaemia
HbA1c	< 5.7%	≥ 6.5% = diabetes	Glycaemic control over ~3 months
ALT / AST	< 40 U/L	> 10× ULN	Hepatocellular injury
ALP	40–130 U/L	—	Cholestasis / bone
Bilirubin (total)	0.3–1.2 mg/dL	—	Jaundice if elevated
Troponin I	< 0.04 ng/mL	> 0.4 = significant	Serial; correlate with ECG
CRP	< 5 mg/L	Marked rise	Inflammation / infection
Procalcitonin	< 0.5 ng/mL	> 2 ng/mL	Suggests bacterial sepsis

3. Urine Analysis Interpretation

Dipstick Finding	Interpretation	Next Step
Protein +	Possible renal disease / UTI	Quantify (ACR); repeat
Glucose +	Hyperglycaemia / low renal threshold	Check blood glucose
Blood +	Haematuria — infection, stones, malignancy	Microscopy; imaging if persistent
Nitrites +	Gram-negative bacteriuria	Send culture; treat if symptomatic
Leucocytes +	Pyuria / inflammation	Correlate with nitrites & symptoms
Nitrites + & Leucocytes +	Strongly suggests UTI	Treat per antibiogram

4. ECG Interpretation Quick Guide

Parameter	Normal	Abnormal Significance
Rate	60–100 bpm	< 60 brady; > 100 tachy
Rhythm	Sinus	AF, flutter, VT, SVT — classify
P-R interval	120–200 ms	> 200 ms = 1° AV block
QRS duration	< 120 ms	> 120 ms = bundle branch block
QTc	< 440 ms (M), < 460 ms (F)	Prolonged → arrhythmia risk
ST elevation	Isoelectric	> 1 mm in ≥ 2 contiguous leads = STEMI protocol

Common arrhythmias: AF — irregularly irregular, absent P waves; **VT** — broad-complex, regular, > 100 bpm, medical emergency; **SVT** — narrow-complex tachycardia, often regular.

5. Arterial Blood Gas (ABG) Interpretation

Parameter	Normal Range	Interpretation
pH	7.35–7.45	< 7.35 acidosis; > 7.45 alkalosis
$PaCO_2$	35–45 mmHg	Respiratory component
HCO_3^-	22–26 mEq/L	Metabolic component
PaO_2	80–100 mmHg	< 60 = respiratory failure
Lactate	0.5–2.2 mmol/L	> 4 suggests hypoperfusion / sepsis

Quick approach: assess pH → identify primary disorder (respiratory vs metabolic) from $PaCO_2$ and HCO_3^- → check for compensation → calculate the anion gap for metabolic acidosis.

6. Thyroid & Cardiac/Sepsis Markers

Test	Normal Range	Note
TSH	0.4–4.0 mIU/L	First-line thyroid screen
Free T4	0.8–1.8 ng/dL	Confirms thyroid status
BNP / NT-proBNP	Age-dependent	Elevated in heart failure
CK-MB	< 5 ng/mL	Cardiac injury (older marker)
Lactate (sepsis)	< 2 mmol/L	≥ 2 triggers sepsis-6 review

7. Radiology Reporting Flags

Modality	Urgent means	Routine TAT
Chest X-ray (CXR)	Pneumothorax, free air, mis-placed line	Reported within 24 h
CT head	Bleed, mass effect, acute infarct	Reported within 12 h
USS abdomen	Free fluid, obstruction, aneurysm	Reported within 24 h

- **Urgent radiology turnaround:** verbal/critical alert within 1 hour of acquisition.
- Reports are viewed on the **PACS** system; clinicians acknowledge critical reports electronically.

8. Microbiology & Culture Turnaround

Specimen / Test	Preliminary	Final Report
Blood culture	Gram stain at flagging (~12–24 h)	48–72 h (5 days if negative)
Urine culture	—	24–48 h
Sputum culture	Gram stain same day	48–72 h
CSF analysis	Urgent Gram stain < 1 h	Culture 48–72 h
Stool culture	—	48–72 h

Antibiotic sensitivities accompany the final report; clinicians should **de-escalate** empirical therapy once the antibiogram is available.

9. Tumour Markers (interpret with caution)

Marker	Associated With	Note
PSA	Prostate	Age-specific; not for screening alone

Marker	Associated With	Note
CA-125	Ovarian	Raised in many benign conditions
CEA	Colorectal	Trend more useful than single value
AFP	Hepatocellular / germ cell	Correlate with imaging

10. Critical Value Notification Protocol

Critical values
The laboratory must telephone the treating ward/doctor **within 15 minutes** of a critical value. The doctor must acknowledge and **document the action within 30 minutes**. If the doctor is unreachable, the lab escalates to the duty registrar, then the on-call consultant.

Document ref: LAB-DIAG-007 · Version 6.1 · Owner: Central Laboratory & Radiology